

RAROTONGA, Cook Islands

WMO: 918430

Lat: 21.2009S

Lon: 159.8130W

Elev: 6

StdP: 101.26

Time Zone: -10.00 (W10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.2	18.2	11.0	8.1	20.5	11.9	8.7	20.6	11.2	22.6	10.0	23.0	1.7	160	0.355

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.8	30.8	26.8	30.1	26.2	29.5	25.8	27.4	29.6	26.8	29.0	26.5	28.6	3.8	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.8	22.4	29.4	26.2	21.6	28.6	25.9	21.2	28.3	87.0	29.6	84.5	29.4	82.8	28.5	32.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.2	8.9	7.9	15.3	32.3	1.0	1.1	14.6	33.1	14.0	33.7	13.5	34.3	12.8	35.1
			13.1	29.1	0.9	1.2	12.5	29.9	11.9	30.6	11.4	31.2	10.8	32.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.7	26.5	26.8	26.8	25.9	24.5	23.4	22.7	22.5	23.0	23.7	24.9	25.7
	DBStd	2.09	1.11	1.12	1.12	1.28	1.49	1.59	1.61	1.62	1.61	1.44	1.37	1.24
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5363	511	471	522	478	449	403	394	388	390	424	448	486
	CDD18.3	2321	252	237	263	228	191	153	136	129	140	166	198	228
	CDH23.3	15712	2310	2289	2498	1823	1095	596	427	381	477	736	1306	1773
(14) Wind	WSAvg	3.9	3.9	4.0	3.7	3.8	3.7	3.8	3.9	3.9	4.2	4.0	3.8	4.1
	PrecAvg	2176	262	228	214	229	196	116	101	136	127	115	161	225
	PrecMax	3924	1099	630	475	566	693	306	250	475	362	333	458	673
	PrecMin	1186	24	12	13	31	18	8	24	19	17	10	20	6
	PrecStd	612	185	116	124	113	140	81	55	95	86	76	117	155
	DB	31.0	31.9	31.8	30.9	29.9	28.9	28.1	27.9	28.2	28.9	30.1	30.8	
	MCWB	27.0	27.7	27.4	27.0	26.3	25.5	24.6	24.2	24.6	25.1	25.9	26.5	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	30.1	30.4	30.5	29.9	28.2	27.6	26.9	26.8	27.1	27.9	29.0	29.4	
	2% MCWB	26.2	26.4	26.6	26.1	24.8	24.3	23.7	23.5	23.9	24.2	24.9	25.5	
	5% DB	29.2	29.9	29.9	29.1	27.8	26.8	26.1	26.0	26.2	27.0	28.2	28.9	
	5% MCWB	25.6	26.0	26.2	25.5	24.4	23.7	22.9	22.8	23.0	23.5	24.4	25.1	
	10% DB	28.9	29.1	29.1	28.3	27.0	26.0	25.3	25.2	25.8	26.2	27.7	28.1	
	10% MCWB	25.3	25.6	25.7	25.0	23.8	23.1	22.4	22.2	22.6	22.7	24.0	24.5	
	WB	27.7	28.4	28.1	27.6	26.7	26.0	25.4	25.2	25.6	26.1	26.6	27.3	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	29.8	30.7	30.6	29.8	28.9	28.0	27.1	26.9	27.3	27.8	28.8	29.5	
	2% WB	26.8	27.2	27.2	26.7	25.9	25.1	24.5	24.3	24.6	25.1	25.8	26.4	
	2% MCDB	28.9	29.4	29.5	28.7	27.6	26.9	26.1	26.0	26.4	26.9	28.0	28.5	
	5% WB	26.3	26.6	26.7	26.2	25.2	24.4	23.9	23.7	24.0	24.5	25.3	25.9	
	5% MCDB	28.5	28.8	28.9	28.3	26.9	26.2	25.5	25.4	25.8	26.3	27.4	27.9	
	10% WB	25.9	26.2	26.2	25.7	24.6	23.7	23.2	23.0	23.3	23.7	24.7	25.4	
	10% MCDB	28.2	28.4	28.4	27.8	26.4	25.5	25.0	24.9	25.3	25.6	26.8	27.5	
(35) Mean Daily Temperature Range	MDBR	4.6	4.8	4.8	4.7	4.8	4.7	4.8	5.0	4.8	4.9	5.0	4.7	
	5% DB	5.1	5.4	5.5	5.5	5.5	5.3	5.4	5.4	5.1	5.3	5.4	5.4	
	5% MCWBR	3.1	3.2	3.2	3.3	3.4	3.4	3.7	3.6	3.3	3.3	3.2	3.1	
	5% WB	4.7	4.9	5.1	5.2	4.8	4.7	4.6	4.9	4.7	4.7	4.9	4.9	
	5% MCWBR	3.1	3.3	3.3	3.5	3.5	3.5	3.9	3.9	3.6	3.4	3.3	3.3	
(40) Clear-Sky Solar Irradiance	taub	0.385	0.384	0.375	0.379	0.373	0.365	0.362	0.361	0.381	0.375	0.377	0.382	
	taud	2.612	2.616	2.650	2.633	2.626	2.628	2.625	2.622	2.563	2.577	2.598	2.598	
	Ebn at Noon	960	948	932	888	855	843	857	894	912	945	960	963	
	Edh at Noon	104	101	94	89	84	80	83	89	101	104	104	105	
(44) All-Sky Solar Radiation	RadAvg	6.34	6.14	5.55	4.71	3.84	3.58	3.78	4.49	5.32	5.99	6.35	6.38	
	RadStd	0.51	0.48	0.41	0.30	0.34	0.28	0.25	0.25	0.33	0.46	0.50	0.52	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.25	N/A	N/A	+0.30	N/A	N/A	N/A	N/A	+124	+123
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page